Measuring What We Promise: A Cross-Institutional Analysis of Learning Outcome Quality in Irish Higher Education

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Abstract

Learning outcomes (LOs) are the primary accountability instrument in Irish higher education, underpinning programme approval, Quality and Qualifications Ireland (QQI) compliance, and the National Framework of Qualifications (NFQ). Yet the quality of LO language — specifically the observability and cognitive demand of the action verbs employed — has received little systematic empirical attention at scale. This paper reports on a corpus-scale analysis of **233,005** module-level LOs drawn from **sixteen** Irish higher education institutions (HEIs), spanning the public university, technological university (TU), Institute of Technology (IT), College of Education, and private/specialised sectors. Using a rule-based natural language processing (NLP) pipeline aligned to a six-dimension evaluation rubric (D1–D6), we score each LO on verb observability (D1), behavioural singularity (D2), and student-centredness (D3). We report a consistent and statistically meaningful gap between the public university and TU clusters on the D1 (verb observability) dimension: public university mean D1 = 1.572 versus TU mean D1 = 1.775 ($\Delta = 0.203$), with public universities showing substantially higher rates of unobservable Tier 0 verbs (8.3% versus 3.4%). We extend the analysis to constructive alignment, cross-tabulating verb tier against assessment modality and, for three institutions with Akari curriculum management data, against LO-to-component mappings, identifying 64 modules in which high-demand LOs are assessed exclusively by formal examination. We discuss implications for institutional quality assurance processes, module approval workflows, and national policy. Preliminary findings from Universal Design for Learning (UDL) proxy analysis and HEA Data and Digital Skills module tagging are also presented.

Keywords: learning outcomes; higher education quality; natural language processing; constructive alignment; verb taxonomy; Irish higher education; QQI; NFQ; Universal Design for Learning

1 Introduction

The learning outcome has become the dominant currency of higher education quality assurance in Europe and Ireland. Across national qualifications frameworks, programme approval processes, institutional review, and external examiner reports, the question asked — implicitly or explicitly — is the same: what can a graduate *do*? The shift from input-based to outcome-based education is now decades old (Adam, 2004; Kennedy, 2007), yet a persistent tension remains between the aspiration embedded in that question and the language used to answer it.

If a module learning outcome states that a student will *understand* the principles of thermodynamics, or will *appreciate* the historical context of modernism, what exactly has been promised? The verb carries the epistemic weight of the outcome statement, yet verbs of this kind — vague, unobservable, unmeasurable — remain prevalent in published module descriptors across the sector (Hussey and Smith, 2002; Moon, 2002). They satisfy the formal requirement of outcome-based language while resisting the accountability that language was designed to enable.

This paper addresses that gap directly. We report on a systematic, corpus-scale analysis of module-level learning outcomes drawn from fourteen Irish HEIs: six public universities, three technological universities, two Institutes of Technology, one College of Education, and two private/specialised colleges. Our corpus of 229,037 LOs was assembled through systematic retrieval of publicly accessible institutional module catalogues, processed through a rule-based NLP pipeline, and evaluated against a six-dimension rubric developed from the literature on LO quality, Bloom's revised taxonomy (Anderson et al., 2001), and the Irish National Framework of Qualifications (Quality and Qualifications Ireland, 2020).

The research makes three substantive contributions. First, it provides the first cross-institutional, corpus-scale empirical characterisation of LO language quality in Irish higher education. Second, it identifies a consistent and reproducible gap between the university and technological university clusters on the dimension of verb observability. Third, it extends the analysis to constructive alignment (Biggs and Tang, 2011), demonstrating that verb quality alone is insufficient: high-demand LOs assessed exclusively by traditional formal examination represent a distinct and actionable category of misalignment.

The paper is structured as follows. Section 2 reviews the relevant literature. Section 4 describes corpus construction, the evaluation rubric, and the scoring pipeline. Section 5 presents findings across all nine institutions with particular focus on the university–TU comparison and constructive alignment. Section 6 interprets findings in the context of Irish HE policy and institutional practice. Section 7 offers recommendations.

2 Background and Related Literature

2.1 Learning Outcomes in Higher Education Policy

The outcomes-based approach to curriculum design entered Irish higher education formally through the Bologna Process and the establishment of the National Framework of Qualifications in 2003 (National Qualifications Authority of Ireland, 2003). QQI's *Policies, Criteria and Guidelines for the Validation of Programmes of Education and Training* (Quality and Qualifications Ireland, 2021) require that all accredited programmes express intended learning outcomes that are specific, assessable, and aligned to NFQ award standards. The National Forum for the Enhancement of Teaching and Learning has further endorsed outcome-based design as a cornerstone of curriculum quality (National Forum for the Enhancement of Teaching and Learning in Higher Education, 2016).

Despite this policy consensus, empirical evidence on the *quality* of LO language in practice is sparse. Institutional review bodies assess whether outcomes are present; they do not routinely measure their linguistic or cognitive quality. Hussey and Smith (2002) identified early that outcome statements frequently conflate multiple behaviours, use unobservable language, or are pitched at levels inconsistent with their NFQ designation. Moon (2002) similarly noted the gap between outcome-as-written and outcome-as-assessable. More recently, Donnelly and Fitzmaurice (2019) observed that Irish HEIs have made significant structural progress in outcome adoption but that the quality of individual statements remains uneven.

2.2 The Policy Framework for Learning Outcomes: Three Levels

*[Note: **[PLACEHOLDER: to be developed with Lisa O'Regan, Centre for Teaching and Learning, MU. See accompanying policy reference document.]**]*

LO requirements for Irish HEIs operate at three intersecting levels. At the *European* level, the Standards and Guidelines for Quality Assurance in the European Higher Education Area (ESG 2015) (European Association for Quality Assurance in Higher Education, 2015) require under Standard 1.2 that programmes are "designed so that they meet the objectives set for them, including the intended learning outcomes" and that "the qualification resulting from a programme should be clearly specified and communicated, and refer to the correct level of the national qualifications framework." The Dublin Descriptors, which underpin the Bologna Process qualification cycles, specify the knowledge, skills, judgement, communication, and learning-to-learn capabilities expected at each cycle, and implicitly require LO language that reaches the cognitive demand appropriate to the cycle (Bachelor, Master, Doctorate) (Joint Quality Initiative, 2004).

At the *national* level, QQI's *Policies, Criteria and Guidelines for the Validation of Programmes of Education and Training* (Quality and Qualifications Ireland, 2021) require that learning outcomes are "specific and sufficiently detailed to communicate the award-holder's knowledge, skill and competence" and that they are "appropriate to the specified NFQ level and award type." The NFQ Grid of Level Indicators defines expected knowledge, skills, and competence at each level; at Level 8 (Honours Bachelor) this includes "mastery of a complex and specialised area of skills and tools" and the capacity to "conduct guided research" — language that maps to observable, high-demand verbs in the D1 Tier 2–3 range. QQI further requires that "assessment procedures should validly and reliably ensure that learners who are recommended for awards have attained the provider's minimum intended programme learning outcomes" — a direct statement of the constructive alignment requirement (Quality and Qualifications Ireland, 2021). The National Forum for the Enhancement of Teaching and Learning has further endorsed outcome-based design and multi-modal assessment (OF, FOR, and AS learning) as cornerstones of curriculum quality (National Forum for the Enhancement of Teaching and Learning in Higher Education, 2016).

At the *institutional* level, each HEI is required under QQI's quality assurance framework

to maintain internal procedures for programme and module design and approval that operationalise these requirements. The gap between this procedural requirement and the documented outcome quality revealed by corpus analysis is a central concern of this study.

Critically, the policy framework does not mandate Bloom's taxonomy by name. It does, however, require precisely what Bloom formalised: outcomes that are specific, observable, level-appropriate, and assessable. Our D1–D6 rubric operationalises the policy criteria without depending on any single pedagogical framework: D1 (verb observability) and D2 (behavioural singularity) address the "specific and sufficiently detailed" requirement; D3 (student-centredness) addresses the student-centred LO format implied by the Bologna student-learning paradigm; D6 (NFQ level calibration) addresses the "appropriate to the specified NFQ level" requirement. The rubric is, in this sense, a policy-compliance instrument as much as a pedagogical quality instrument.

[TODO: Lisa O'Regan to develop: (a) institutional policy context at MU (L&T Strategy, Programme Board procedures); (b) national QA framework developments post-2021; (c) comparative European context — how does Irish policy implementation compare with EHEA peers? (d) implications of the shadow curriculum problem for institutional QA: if good practice goes undocumented, internal approval processes cannot credit or require it.]

2.3 Bloom's Taxonomy and Verb Observability

The most widely applied framework for evaluating the cognitive demand of learning outcomes is Bloom's Taxonomy of Educational Objectives (Bloom, 1956) and its subsequent revision by Anderson et al. (2001). The revised taxonomy distinguishes six levels of cognitive process — Remember, Understand, Apply, Analyse, Evaluate, Create — each associated with a set of observable action verbs. The verb is the operationalisation of the level: it specifies what the student must *do* and, by extension, what the assessor must observe.

The critical distinction for quality assurance purposes is between verbs that are *observable* — whose instantiation can be directly evidenced in student performance — and those that are *unobservable*. Verbs such as *understand*, *know*, *appreciate*, and *be aware of* describe internal cognitive states that cannot be directly measured. Their prevalence in module descriptors signals either weak outcome design or a disconnect between the language

of course documentation and the reality of teaching and assessment (Kennedy, 2007). Importantly, the problem is not that deep understanding is an unworthy goal — it is that the verb provides no specification of what evidence of understanding would look like.

2.4 Constructive Alignment

Biggs's constructive alignment framework (Biggs, 1996; Biggs and Tang, 2011) provides the theoretical basis for our extension of the analysis beyond verb quality. Alignment requires internal consistency across three elements: intended learning outcomes, teaching and learning activities, and assessment tasks. A module in which the LO verbs are pitched at high cognitive demand (e.g., *design*, *evaluate*, *synthesise*) but the sole assessment instrument is a closed-book formal examination represents a specific form of misalignment: the assessment cannot elicit the performance the outcome specifies.

This form of misalignment — high-demand LOs, low-authenticity assessment — is practically significant. It disadvantages students who possess the higher-order capability described in the outcome but whose performance is measured by a proxy (recall under examination conditions) that conflates memory with competence. It also undermines the validity of grades as signals of graduate capability, with consequences for employers, professional bodies, and the integrity of NFQ level claims (Bloxham and Boyd, 2007).

2.5 Automated Analysis of Learning Outcome Text

The application of NLP methods to educational text is a growing field (Demetriadis et al., 2018; Zawacki-Richter et al., 2019). Prior work has applied automated methods to the classification of educational objectives (Khandelwal et al., 2021; Ullmann, 2019), the detection of LO quality issues (Chen et al., 2022), and cross-programme curriculum mapping (Shabani et al., 2012). However, corpus-scale cross-institutional analysis of LO language quality — spanning an entire national HE system — has not previously been reported.

The availability of machine-readable module catalogues across Irish institutions, combined with advances in rule-based and large language model (LLM) classification, makes such analysis technically feasible at scale for the first time. Our study exploits this opportunity.

3 Research Questions

This study addresses three research questions:

RQ1 What is the distribution of verb observability tiers across module-level learning outcomes in Irish higher education, and does this distribution differ systematically between the university and technological university sectors?

RQ2 What is the prevalence of constructive misalignment — defined as high-demand LOs paired with assessment modalities that cannot elicit the specified performance — across the corpus?

RQ3 What methodological and policy implications follow from corpus-scale LO quality analysis for institutional module approval processes and national quality assurance frameworks?

4 Methodology

4.1 Corpus Construction

Module learning outcome data were collected from sixteen Irish HEIs through systematic automated retrieval of publicly accessible module catalogue pages during the period October 2024 – April 2026. The institutions comprised six public universities (Maynooth University [MU], University College Dublin [UCD], University College Cork [UCC], University of Galway [Galway], University of Limerick [UL], and Dublin City University [DCU]), three technological universities (Atlantic Technological University [ATU], South East Technological University [SETU], and Munster Technological University [MTU]), two Institutes of Technology (Dundalk Institute of Technology [DkIT] and Dún Laoghaire Institute of Art, Design and Technology [IADT]), three private and specialised colleges (National College of Ireland [NCI], National College of Art and Design [NCAD], and Griffith College Dublin [GC]), one College of Education (Marino Institute of Education [MIE]), and TU Dublin [TUD], whose catalogue format is discussed separately as a methodological finding (Section 5.4).

Sectoral terminology. We use *public universities* (PU) to refer to institutions designated under the Universities Act 1997. The broader Irish HE system also includes Technological Universities (TU), Institutes of Technology (IT), Colleges of Education (CE), and private and specialised colleges (PS). The PU label does not imply that TUs are private or non-public — all Irish HEIs receive public funding. The distinction is legislative and sectoral. **[TODO: Confirm preferred terminology; check HEA System Performance Framework for current administrative categories.]**

Table 1 summarises the corpus. Each institution required a bespoke retrieval approach reflecting the diversity of underlying catalogue systems: CourseLeaf (UCC, Selenium headless Chrome), Akari curriculum management system (MTU, SETU, DkIT, NCI — sequential module-ID sweeps), BookOfModules (UL), the nuigalway.ie course catalogue (Galway), modspec.dcu.ie and dcu.akarisoftware.com (DCU), TU Dublin Vault (TUD), and institution-specific HTML catalogues for the remaining institutions. IADT and NCAD module data were extracted from publicly available PDF catalogues using pdfplumber. MIE data were parsed from four programme handbooks. In all cases, module-level data were extracted including: module code, title, ECTS credits, NFQ level where published, and the full text of each numbered learning outcome.

Table 1: Corpus composition by institution (D1–D3 scoring complete for all).

Institution	**Type**	**Modules**	**LOs**	**D1 Mean**	**Tier 0 (%)**
MU	PU	2,139	12,603	1.637	10.0
DCU	PU	3,464	14,472	1.682	9.5
Galway	PU	2,847	12,477	1.627	9.7
UCC	PU	6,802	29,102	1.697	4.9
UCD	PU	5,419	27,145	1.158	13.4
UL	PU	5,731	27,056	1.736	4.9
PU cluster		*26,402*	*122,855*	*1.572*	*8.3*
ATU	TU	5,949	33,770	1.846	5.0
SETU	TU	3,772	18,162	1.529	1.6
MTU	TU	8,217	29,881	1.846	2.8
TU cluster		*17,938*	*81,813*	*1.775*	*3.4*
DkIT	IT	4,697	19,590	1.644	6.6
IADT	IT	175	814	1.776	4.9
NCI	PS	749	3,647	1.627	10.4
NCAD	PS	79	269	1.796	2.6
GC	PS	63	441	1.914	0.5
MIE	CE	163	49	1.592	0.0
Total		**51,190**	**233,005**	—	—

ATU: cross-programme deduplication. ATU LO data were retrieved from programme pages on `atu.ie`, where modules appear in multiple degree programmes. Raw extraction yielded 69,532 LOs across a stated 14,200 module instances; after removing cross-programme duplicates (deduplication key: module code × LO index × normalised text) the corpus reduces to 33,770 unique LOs from 5,949 unique module codes — a 51.4% reduction. A further 622 records are retained as shared-code variants (same module code, different content and credit loading), identified by a version suffix in their identifier.

DCU: Institute of Education modules. The DCU corpus includes modules from the DCU Institute of Education, which incorporates the former Church of Ireland College of Education (CICE), St Patrick's College Drumcondra, and Mater Dei Institute (all merged into DCU in 2016).

UL: Mary Immaculate College. The UL corpus includes modules validated under UL's degree-awarding authority for Mary Immaculate College (MIC), Limerick. MIC-specific subject codes (MH, GY, RS) are confirmed present; B.Ed and PME modules are integrated under shared UL subject codes and cannot be disaggregated without department-level metadata.

MIE: programme handbook extraction. MIE data were extracted from four publicly available programme handbooks (B.Ed. Primary, B.Sc. ECE, B.Sc. Ed. Studies, B.Oid.). Of 163 modules across the four programmes, only 14 (8.6%) contain explicit student-centred learning outcomes in the handbook text; the remainder provide module descriptions only. The B.Sc. Ed. Studies handbook contains no module-level LO statements. D1 mean and Tier 0 are computed from the 49 English-language LOs only. The B.Oid. (Irish-medium B.Ed.) handbook contains module descriptions in Irish (Gaeilge); these 38 modules are included in the module count but excluded from D1 scoring.

Griffith College: PDF extraction. GC module LO data were extracted from 74 PDF module descriptor documents retrieved from Griffith College programme pages. Each PDF follows a standardised format with module title, ECTS credits, NFQ level, and numbered learning outcomes. 63 of 74 PDFs contained parseable LO lists; the remainder provided aims-only descriptors and were excluded. No module codes are published in these PDFs; GC records carry no `moduleCode` field.

TU Dublin: dual-source retrieval. TU Dublin was surveyed via two routes. The primary public catalogue (Vault system, 5,449 module pages) does not publish student-centred LOs; modules use an aims format (*"The aim of this module is to [verb]. . . "*). PDF module catalogues published by two TUD schools were retrieved separately and yielded 120 modules and 736 student-centred LOs across six programmes (TU835 Architecture,

TU874 Mathematical Sciences, TU078, TU079, TU248, TU5294). The combined TUD corpus (3,527 records) is included in the totals; the aims-vs-LO distinction is discussed as a methodological finding in Section 5.4.

DkIT and IADT. Dundalk Institute of Technology (DkIT) contributes 4,697 modules and 19,590 LOs (D1 mean 1.644). Its D1 mean is notably lower than the other TU-sector institutions and closer to the university cluster, which may reflect its status as a remaining IoT rather than a designated TU; this is discussed in Section 6. IADT module descriptors were retrieved from a single publicly accessible PDF module catalogue (175 modules, 814 LOs, D1 mean 1.776). No assessment data are published in this document; IADT contributes to LO quality analysis only.

Excluded institutions. Trinity College Dublin (TCD) is under active investigation but is excluded from the present analysis due to its decentralised catalogue structure. TUS (Technological University of the Shannon: Midlands Midwest) and RCSI University of Medicine and Health Sciences are also not yet included.

4.2 Module Descriptor Enrichment

Each LO record in the corpus was augmented with its module descriptor — the contextual summary text (typically 50–300 words) that precedes the LO list in the published module catalogue. Descriptors provide the interpretive context needed for LLM-based D4–D6 scoring and for the accompanying LO Browser application (Section **??**). Descriptor retrieval required institution-specific strategies, summarised in Table 2.

Table 2: Module descriptor enrichment by institution.

Institution	Source / Method	M
MU	Institutional data export (`moduleContent` field)	
ATU	`moduleContent` field in LO records (per-LO text)	
UCC	`ucc-modules-raw.json` description field	
UL	`ul-modules-raw.json` description field	
MTU	`mtu-modules-raw.json` description field	
TUD	`tud-modules-raw.json` description field	
SETU	Akari module records description field	
DkIT	Akari module records description field	
UCD	`ucd-modules-raw.json` description field	
Galway	Scraped from `nuigalway.ie/course-information/`	
DCU	modspec.dcu.ie (old and Akari endpoints); dcu.akarisoftware.com	
NCI	Scraped from `nci.akarisoftware.com` (indicative content)	
NCAD	Extracted from PDF module descriptors (pdfplumber, Introduction section)	
IADT	Extracted from PDF module catalogue (pdfplumber, aims section)	
GC	Extracted from PDF module descriptors (pdfplumber, Module Objectives section)	
MIE	Not available (programme handbooks contain no descriptor text)	

[a] UCD's catalogue does not publish descriptors for all modules; coverage is limited by institutional catalogue completeness.

[b] DCU descriptor coverage reflects an incomplete migration to the Coursebuilder/Akari system at time of collection; modules not yet migrated have no machine-readable descriptor in any publicly accessible endpoint.

Descriptor text was extracted using institution-appropriate methods: direct field reads from JSON catalogue exports (MU, ATU, UCC, UL, MTU, TUD, SETU, DkIT, UCD); HTTP scraping with targeted HTML parsing (Galway, NCI, DCU); and PDF text extraction using `pdfplumber` with section-boundary detection based on heading patterns (NCAD: "1. Introduction" section; IADT: aims text preceding the "On successful completion" LO trigger; GC: "Module Objectives" or "Module aims and objectives" section). Non-English descriptors (primarily Irish-medium MIE and UL/MIC modules) are retained without language filtering for subsequent analysis. Descriptor text is stored in the `moduleContent`

field of each normalised LO record and is available through the LO Browser application.

4.3 A Tripartite Evaluation Framework: Measurement, Evaluation, and Validation

A central methodological premise of this study is that LO quality analysis requires two epistemologically distinct operations that must be kept separate: *measurement* and *evaluation*. These are not synonyms, and conflating them produces analytically incoherent results.

Measurement (the NLP pipeline) applies deterministic rules to observable surface features. It *calculates*: given an LO string, the D1 score is computable without interpretation, context, or reference to institutional intent. The same algorithm applied to the same text will always produce the same result. This reproducibility is a strength for corpus-scale analysis but a limitation for quality assessment: a rule-based system cannot read an LO against the policy standards it is meant to satisfy, cannot judge whether a verb choice is appropriate for a given NFQ level and credit load, and cannot recognise when a syntactically well-formed LO is substantively empty.

Evaluation (the LLM and human raters) applies normative judgement to contextualised instances. It *assesses*: given an LO, its module context, its institutional setting, and the policy framework the institution is required to meet, a rater determines whether the outcome constitutes a quality statement. Evaluation is inherently interpretive. This is both a limitation (rater effects, anchoring, disciplinary bias) and a capability (sensitivity to context, policy grounding, and the kind of reasoning that expert quality reviewers actually apply).

The research question that the LLM and human raters are answering is:

> *Does this learning outcome, read in context and against the standards its institution is required to meet, constitute a well-formed, assessable, and level-appropriate outcome?*

This is the question a QQI validation panel would ask. It is not reducible to a calculation.

The third stream — **validation via inter-rater reliability (IRR)** — is the mechanism

by which the measurement and evaluation streams are brought into dialogue. By computing pairwise agreement between the NLP pipeline, the LLM evaluator, and expert human raters on a stratified sample, we can determine: (a) the extent to which automated measurement captures the same variance as expert evaluation; (b) where the two approaches diverge and why; and (c) whether policy-grounded LLM evaluation achieves sufficient agreement with human judgement to serve as a scalable quality audit instrument.

4.4 The D1–D6 Evaluation Rubric

We developed a six-dimension evaluation rubric (Table 3) grounded in the LO quality literature and calibrated to the Irish NFQ context. Each dimension is scored 0–3, yielding a composite score of 0–18. D1–D3 are operationalised through the rule-based NLP pipeline (measurement); D4–D6 require contextual judgement and are operationalised through the LLM evaluator and human raters (evaluation).

Table 3: The D1–D6 evaluation rubric and the tripartite framework.

#	Dimension	Description	Stream
D1	Verb observability	Cognitive demand and observability of the leading action verb	NLP (measurement)
D2	Behavioural singularity	Degree to which the LO specifies a single, bounded behaviour	NLP (measurement)
D3	Student-centredness	Grammatical framing from the student's perspective	NLP (measurement)
D4	Scope appropriateness	Whether the scope matches NFQ level and ECTS credit load	LLM + Human (evaluation)
D5	Assessability signal	Extent to which the LO implies a specific assessment approach	LLM + Human (evaluation)
D6	NFQ level calibration	Alignment of cognitive demand to the stated NFQ level	LLM + Human (evaluation)

For IRR purposes, the LLM evaluator also independently scores D1–D3 on the same sample, providing a cross-validation signal between the measurement and evaluation streams on the dimensions where both can operate.

4.5 NLP Pipeline: Overview and Pre-processing

The rule-based NLP pipeline is implemented in a single-pass Node.js scorer (`score-rule-based.mjs`) that reads normalised LO records and produces per-LO scores for D1–D3 together with structured rationale strings and diagnostic flags. The pipeline is deterministic and reproducible: given the same LO text and lexicon version, it returns identical output on every

run.

Input normalisation. Each raw LO text undergoes a cleaning pass before analysis:

1. *Numbering prefix removal.* LO text extracted from PDFs and web pages frequently retains the source numbering (`1.`, `Lo 1.`, `i.`, `(a)`). A family of regular expressions strips these prefixes before any linguistic analysis.

2. *Language detection.* A word-list of high-frequency Irish (Gaeilge) function words and verbal particles is checked against the LO text. Irish-medium LOs (`language=ga`) are excluded from D1–D3 scoring and assigned a `null` score with an explanatory flag. This affected 33 modules from MIE's B.Oid. programme; the B.Oid. modules are retained in the corpus with their metadata intact.

3. *Infinitive and framing prefix stripping.* The leading verb is isolated by removing common preamble patterns before attempting verb extraction: *"To be able to"*, *"To"* (bare infinitive), *"Be able to"*, and third-person *"Students will (be able to)"* wrappers are stripped so that the true action verb is presented for lookup.

Diagnostic flags. The pipeline produces a structured `flags` array per LO for quality audit purposes. Flags include: `unknown-verb` (verb not in lexicon, requiring review); `no-action-verb` (LO opens with a noun or adjective rather than an action verb, e.g. *"Understanding of... "*, *"Knowledge of... "*); `wrong-tense` (past-tense or participial form rather than imperative, e.g. *"Demonstrated... "*, *"Applying... "*); `irish-language`; `compound` (multiple behaviours detected); `instructor-centered`; and `context-dependent` (verb whose observability tier is context-sensitive). LOs carrying a null D1 score are excluded from mean calculations but retained in the corpus.

4.6 D1 Scoring: Verb Observability Tier

D1 operationalises the observability of the leading action verb, grounded in Mager's (1962) criterion of behavioural observability and the revised Bloom's taxonomy (Anderson et al., 2001). The central instrument is a tiered verb lexicon (v1.3) developed iteratively from the corpus and reviewed against published LO quality guides from QQI 2017b, the ECTS

Users' Guide 2015, and the Cedefop 2017 handbook on defining and writing learning outcomes — the document explicitly cited by QQI as the normative reference for LO writing guidance.

Lexicon structure. The lexicon (367 entries at v1.3, distributed as supplementary material) assigns each verb to one of four tiers:

Tier 0 — Unobservable (50 verbs) Internal cognitive states that cannot be directly assessed without further specification: *understand, know, appreciate, be aware of, learn, grasp, realise, be familiar with, internalise.* These map to Mager's *fuzzy* verbs and correspond to Anderson and Krathwohl's *Remember* and undifferentiated *Understand* categories where no observable performance is implied.

Tier 1 — Weak observable (53 verbs) Behaviours that imply an observable output but are insufficiently specific to determine the assessment method without additional context: *describe, identify, discuss, explain, list, name, state, define, outline, summarise.*

Tier 2 — Clear observable (207 verbs) Behaviours with a defined performance criterion and a clear assessment product: *analyse, evaluate, apply, compare, contrast, demonstrate, classify, interpret, solve, justify, differentiate, predict.*

Tier 3 — Precise/contextual (57 verbs) High-demand behaviours with strong domain specificity implying both the action and its expected artefact or product: *design, construct, synthesise, critique, formulate, derive, implement (novel solution), develop (original), compose, engineer.*

Multi-word verbs (*be aware of*, *critically evaluate*, 18 in total) are matched using a longest-match greedy strategy before single-token lookup. Eleven verbs are flagged as *context-dependent*: their observability tier depends on the domain or the surrounding noun phrase in ways that a lexical lookup cannot resolve (*understand* is the canonical example, but *reflect*, *investigate*, *communicate*, and *implement* also appear in this category). Context-dependent verbs are scored by their primary-tier assignment and flagged for potential LLM re-evaluation in the D6 pass.

Verb extraction algorithm. After pre-processing strips leading preamble, the following extraction sequence is applied:

1. *Multi-word greedy match.* All multi-word lexicon entries (sorted longest-first) are checked; the first match wins.

2. *"have [past-participle]"* forms. Some LOs use perfect aspect (*"have identified. . . "*); the auxiliary is discarded and the participle is used as the verb candidate.

3. *"be [adjective]"* competency framing. LOs opening with *"be [adjective]"* (e.g. *"Be competent in. . . "*) describe a state rather than a performance. These receive D1 = 1 (weakly observable state) with a `competency-framing` flag.

4. *Adverb detection.* The qualifier *critically* is treated as a tier-boosting adverb: if the opening token is *critically* and the following verb maps to Tier $t < 3$, the score is elevated to $\min(t + 1, 3)$. Style-only adverbs (*clearly, effectively, confidently*) are stripped without affecting the tier.

5. *Noun/adjective opener detection.* A 32-token set of common noun or adjective openers (*understanding, ability, knowledge, skills, competence, awareness, . . .*) triggers a `no-action-verb` flag and a null D1 score. These represent a recurring structural defect in which an LO describes a capability attribute rather than an observable performance.

6. *Wrong-tense detection.* Past-tense and participial forms (*developed, demonstrated, applying, exploring*) trigger a `wrong-tense` flag and a null D1 score, with a rationale noting the required correction.

7. *Single-token lookup.* The remaining first token is looked up in the verbTier map (case-insensitive). A match returns the tier score; no match returns null with an `unknown-verb` flag.

D1 scoring outcome. Each scoreable LO receives D1 $\in \{0, 1, 2, 3\}$. Null scores (Irish-language, no-action-verb, wrong-tense, unknown-verb) are excluded from institution means but counted separately. Across the corpus, null D1 scores represent approximately 2.1% of all LO records. **[TODO: Insert final null rate from corpus-wide stats.]**

4.7 D2 Scoring: Behavioural Singularity

D2 operationalises the principle that each LO should specify a single bounded, independently assessable student behaviour (Biggs and Tang, 2011; Quality and Qualifications Ireland, 2017a). Compound LOs — those that bundle multiple distinct behaviours into a single statement using coordinating conjunctions — reduce the granularity available for assessment mapping and progression tracking.

Detection algorithm. After stripping the leading infinitive opener (*"To"*), the LO text is split into clauses at the following boundary tokens: *" and "*, *" or "*, semicolons, and commas followed by a space and a lowercase letter (to catch comma-listed verb phrases). For each resulting clause, the first token is extracted and checked against the verb lexicon. Tokens not in the lexicon are not counted. The count of distinct recognised verbs (*not* the raw clause count) determines the D2 score:

D2 score	Criterion
3	Single behaviour — one recognised action verb, no coordination
2	Two distinct behaviours — compound but manageable
1	Three behaviours — conflated; splitting recommended
0	Four or more behaviours — highly fragmented

Limitations. The detection is conservative by design: only tokens present in the verb lexicon are counted, so a compound LO built from unlexiconed verbs may be under-counted. Adversely, a single verb with several qualifying noun phrases (e.g., *"analyse the structure, function, and regulation of..."*) is not falsely scored as compound. D2 is best interpreted as a lower bound on behavioural multiplicity; the LLM pass (D5, assessability signal) is intended to provide a context-sensitive complement.

4.8 D3 Scoring: Student-Centredness

D3 assesses the grammatical framing of the LO, distinguishing LOs that place the student's observable performance at the centre from those framed from the module's or instructor's perspective. The distinction is required by QQI policy 2017a and the Bologna/ECTS

framework 2015, both of which specify that LOs must describe what the *student* will be able to demonstrate upon completion, not what the *module* will cover.

Scoring scheme. Patterns are checked in priority order (first match wins):

D3 = 0 — Instructor/module-centred LO begins with one of 14 instructor-centred patterns: *"This module will... "*, *"The aim of this module is to... "*, *"This course will introduce... "*, etc. These are aims statements masquerading as LOs; the grammatical subject is the module or course, not the student.

D3 = 1 — Passive student LO places the student as a recipient rather than an agent: *"Students will be introduced to... "*, *"Students will be provided with... "*, *"Students will become familiar with... "*. Nine such passive constructions are detected. These are student-centred in framing but describe an input to the student rather than an output from the student.

D3 = 2 — Active student, indirect framing LO states what the student will do, but uses a third-person or infinitive wrapper: *"Students will [verb]... "*, *"The student will be able to... "*, *"To [verb]... "*. These are student-centred and assessable but use a more indirect grammatical structure than the preferred OBE convention.

D3 = 3 — Verb-first imperative LO opens directly with a recognised action verb, placing the student performance as the grammatical subject by implication: *"Analyse... "*, *"Design... "*, *"Demonstrate... "*. This is the preferred convention in outcome-based education (Biggs and Tang, 2011) and is explicitly recommended by QQI 2017a.

Relationship to D1. D3 and D1 are orthogonal by construction: a verb-first LO can score D3 = 3 with a Tier 0 verb (*"Understand the principles of... "*), and a module-centred framing can contain a high-tier verb (*"This module will develop students' ability to synthesise... "*). The two dimensions capture independent quality properties and are reported separately throughout the analysis.

4.9 LLM Evaluation: D4–D6 (Pilot)

The three rule-based dimensions (D1–D3) operate on lexical and syntactic surface features and are context-free by design: the D1 score for *"understand the principles of machine learning"* is the same regardless of whether the module is a first-year introductory unit or a Level 9 specialist course. The remaining three dimensions (D4 scope appropriateness, D5 assessability signal, D6 NFQ calibration) require contextual judgement that is beyond the capacity of rule-based methods, and are therefore operationalised through an LLM evaluator.

Architecture. The evaluator (`score-llm-claude.mjs`) makes a single structured API call per LO to the Anthropic Claude API, using a prompt that combines:

1. **Module context:** module code, title, department, ECTS credits, NFQ level, and the full module descriptor text (truncated to 200 words where necessary).
2. **The LO under evaluation,** identified by its within-module index.
3. **The D4–D6 rubric,** presented as a structured scoring instrument with explicit 0–3 anchor descriptions per dimension.
4. **NFQ level descriptors,** mapping each integer level to the QQI knowledge, skill, and competence expectations for that level.

The model is instructed to respond with valid JSON in a fixed schema (`{d4: {score, rationale}, d5: {score, rationale}, d6: {score, rationale}}`) and to reason step by step before committing to each score. Response temperature is left at the API default (not set); scores are clamped to $[0, 3]$ and rounded to the nearest integer. Responses that do not parse as valid JSON, or that are missing any required dimension, are logged as errors and excluded from analysis.

Current status. A pilot evaluation has been completed on the MU corpus ([**TODO: insert n from scored-llm-claude.json**]), using `claude-haiku-4-5` as the evaluating model. Full cross-institutional evaluation is planned but not yet complete. Results from the pilot are not reported in the current paper; this section documents the methodology for the follow-on analysis.

Policy-grounded prompt architecture. The v2 evaluator (`score-llm-v2.mjs`) extends the pilot with a two-tier prompt structure designed to ground the LLM in the same normative framework as an expert human rater. The **system prompt** (fixed across all calls, eligible for Anthropic prompt caching) contains:

- **QQI Policies and Criteria for the Validation of Programmes** (Quality and Qualifications Ireland, 2017b) and **QQI Core Policies and Criteria** (Quality and Qualifications Ireland, 2024) — the Irish national standards requiring that module LOs are explicitly specified, plainly written, not vague or generalised, and appropriate to NFQ level and credit volume.
- **QQI Assessment and Standards** (Quality and Qualifications Ireland, 2022a) — requiring assessment to be criterion-referenced against stated outcomes.
- **ECTS Users' Guide** (European Commission, 2015) — the Bologna framework requiring demonstrable achievement and discouraging unobservable verbs such as *understand* and *know*.
- **Cedefop (2017)** (Cedefop, 2017) — QQI's stated normative reference for LO writing guidance, including active verb requirements, specificity criteria, and measurability standards.
- **NFQ level descriptors** (levels 6–10) — knowledge, know-how/skill, and competence expectations at each level, used to ground D6 calibration.
- **The D1–D6 rubric**, presented as a structured scoring instrument with explicit 0–3 anchor descriptions per dimension.

The **user prompt** (per LO) contains an institution-specific context block selected by institution code, drawing from a corpus of LO writing guidance documents collected from all sixteen institutions in the corpus. These documents range from full standalone institutional guides (DCU Teaching Enhancement Unit, 2023; UCC CIRTL, 2023) to QA framework appendices and curriculum review checklists. Where an institution has adopted QQI or Kennedy 2007 as its primary reference without supplementary guidance, the evaluator falls back to the shared policy layer. The user prompt additionally provides module code, title, ECTS credits, NFQ level, and module descriptor text.

This architecture means the evaluator is answering the question a QQI validation panel would ask: not merely whether an LO contains an action verb, but whether it meets the standards the institution is itself required to apply.

Computational efficiency: prompt amortisation through batching. A policy-grounded LLM evaluator introduces a structural cost asymmetry: the system prompt — containing the full QQI, ECTS, Cedefop, and NFQ policy layer plus the evaluation rubric — is approximately 1,200 tokens and is fixed per API call, while the per-LO user content (institutional context block, module descriptor, and LO text) is approximately 265 tokens. When LOs are evaluated one per call, the fixed cost dominates: approximately 82% of input tokens are system prompt, applied to a single LO. Sending N LOs per call amortises the system prompt across N instances, reducing the effective input token cost per LO to:

$$C_{\text{input}}(N) = \frac{T_{\text{sys}}}{N} + T_{\text{lo}} \tag{1}$$

where T_{sys} is the system prompt token count and T_{lo} is the per-LO user content token count.

Table 4 shows empirical results from the MU pilot (`gpt-4o-mini`; $n = 50$ per condition; 300 output tokens budgeted per LO).

Table 4: Effect of batch size on LLM evaluation cost and throughput (MU pilot, gpt-4o-mini).

Batch	Input tok/LO	Output tok/LO	Cost/LO	LOs/min	Full corpus est.
1	1,465	162	\$0.000317	14.6	\$73.9
5	516	163	\$0.000175	16.6	\$40.8
10	398	143	\$0.000146	31.3	\$34.0

The input token saving follows the expected curve: moving from batch 1 to batch 5 reduces per-LO input tokens by 65%; moving from batch 5 to batch 10 yields a further 23% reduction, reflecting the diminishing-returns character of the harmonic series. Output tokens are essentially constant across batch sizes (143–163 tokens per LO), since each LO's evaluation is generated independently within the response. Throughput at batch 10

doubles relative to batch 1 (31.3 vs 14.6 LOs/min) because fewer API round-trips are required regardless of per-call latency.

Error rates were zero at both batch 5 and batch 10 when output token budget was set to $300 \times N$ tokens per call. A tighter budget (150 tokens per LO, used in initial testing) produced truncation errors on LOs with long module descriptors; increasing the budget resolved this without materially increasing cost, since output tokens for short responses are not charged for unused budget.

The practical recommendation for corpus-scale evaluation is batch 5 as a conservative choice (well-tested, low error rate, 45% cost reduction vs single-LO) or batch 10 for maximum efficiency (54% cost reduction, doubled throughput, zero errors in pilot). Batches larger than 10 are not recommended: the model must track more simultaneous evaluation targets within a single response, increasing the risk of cross-LO confusion, and the marginal cost saving beyond batch 10 is small.

This batching approach generalises to any policy-grounded LLM evaluation task where the system prompt is substantially larger than the per-instance user content — a common configuration when providing regulatory or institutional context for document quality evaluation.

Inter-rater reliability design. A stratified sample of 300–500 LOs will be drawn from across the corpus (stratified by institution sector, NFQ level, and D1 tier to ensure coverage of edge cases) and evaluated by three raters: (1) the rule-based NLP pipeline (D1–D3); (2) the LLM evaluator (D1–D6, providing an independent D1–D3 signal for cross-validation); and (3) one or more expert human raters. Pairwise inter-rater reliability will be reported as Krippendorff's α for ordinal D1–D6 scales and Cohen's κ for D3 as a binary (student-centred / not student-centred) measure. The IRR analysis is reported in a companion paper. **[TODO: Update when IRR data are available.]**

4.10 Human Evaluation

Human evaluation provides the interpretive anchor against which both the NLP and LLM streams are validated. Expert raters — academic staff with experience in programme design, QQI validation processes, or quality assurance — apply the D1–D6 rubric independently

to a stratified sample of the corpus, with the same policy context documents available to the LLM evaluator.

Human evaluation is not positioned as a superior method but as a different one. It brings disciplinary sensitivity and institutional knowledge that neither the NLP pipeline nor the LLM evaluator can fully replicate; it is also subject to rater fatigue, anchoring effects, and disciplinary bias that automated methods avoid. The three streams are therefore complementary: the NLP pipeline provides scalable, reproducible measurement across the full corpus; the LLM evaluator provides policy-grounded evaluation at scale; and the human raters provide the interpretive anchor that validates both.

Rating tool. To facilitate distributed human evaluation, a browser-based rating interface was developed and deployed at https://jkeatingmu.github.io/MU-finder/slo-study/ (see also Appendix B). The tool is fully static — it requires no login, no plugin, and no software installation — and is accessible to any collaborator with the URL. Raters browse the full 233,005-LO corpus by institution, expand any LO record to view the full module context and NLP/AI evaluation scores, and submit ratings via an in-app modal.

The modal presents the LO text and module context at the top for reference, then solicits scores (0–3) and optional rationale text for each of the four human-stream dimensions: D1 (verb observability, for IRR against the NLP measurement), D4 (scope), D5 (assessability), and D6 (NFQ calibration). Scores are submitted anonymously to a cloud database (Supabase); no personal identifying data are collected or transmitted. The rater's browser records which LOs have been rated in local storage, allowing the interface to indicate previously rated records and warn of potential duplicate submissions; this record is local only and is never transmitted.

IRR sample design. Inter-rater reliability will be computed on a stratified sample of approximately 200–300 LOs, drawn proportionally by institution type (university, technological university, institute), NFQ level (6–10), and D1 tier (0–3) to ensure coverage of the quality gradient. Each LO in the sample will be independently rated by at least two human raters. Agreement will be assessed using Cohen's weighted κ for each dimension, with a target $\kappa \geq 0.70$ as the threshold for acceptable reliability (Landis and Koch, 1977).

The same sample will be evaluated by the LLM pipeline, enabling a three-way comparison: NLP measurement vs. LLM evaluation vs. human evaluation.

Rater protocol. Raters will receive a briefing document presenting the rubric, the underlying policy framework (QQI QP.17, ECTS Users' Guide, Cedefop 2017, NFQ descriptors), and ten calibration LOs with consensus scores and written rationale. Raters are instructed to evaluate each LO as an independent unit — without anchoring to other outcomes within the same module — and to apply the rubric as written rather than adjusting for perceived institutional norms.

[TODO: Human evaluation data collection in progress — IRR results to be reported when minimum sample threshold is reached.]

4.11 Constructive Alignment Analysis

For the constructive alignment extension, we matched module D1 scores against assessment modality data. At Maynooth University, assessment data were extracted from the institutional module descriptor system: each module records the percentage weight of continuous assessment (CA) versus formal examination. At MTU, the Akari assessment management system provides richer data, including per-component assessment type, weight, and — critically — a mapping from each component to the specific LO indices it is designed to assess.

Misalignment was operationalised as follows: a module is classified as **Type A** (confirmed misalignment) when its mean D1 verb tier ≥ 2.0 (clear observable or above), its formal examination weight $\geq 70\%$, and (for MTU) the high-demand LOs are specifically mapped to the examination component with no CA rescue. A module is classified as **partially rescued** when high-demand LOs appear in both the examination and CA component mappings. Type B misalignment (low D1 + high CA weight) captures the converse problem: assessment practice that exceeds the aspiration of the LO language.

5 Results

5.1 Cross-Institutional D1 Distribution

Figure ?? **[TODO: Insert bar chart: D1 mean by institution, coloured by sector (PU/TU/PS)]** shows the D1 mean for each of the ten institutions. The within-PU-sector consistency is notable, with five of six universities returning means between 1.627 and 1.736; UCD (1.158) is an outlier requiring methodological scrutiny (see below). TU means range from 1.529 (SETU) to 1.846 (ATU and MTU), with SETU occupying a position closer to the university cluster than the other TUs — a finding discussed in Section 6. The sector gap of $\Delta = 0.203$ (PU mean 1.572, TU mean 1.775) is consistent in direction across every institution; no TU mean falls below the five-university mean excluding UCD (1.695).

Table 5: D1 tier distribution by sector.

	Tier 0		**Tier 1**		**Tier 2+**	
Sector	**n**	**%**	**n**	**%**	**n**	**%**
University	10,198	8.3	71,256	58.0	41,401	33.7
TU	3,827	3.7	51,243	49.5	48,370	46.8

The Tier 0 rate differential is particularly notable: 8.3% of PU LOs open with an unobservable verb versus 3.4% in the TU sector ($\Delta = 4.9$ percentage points). At UCD, which returns the lowest D1 mean in the corpus (1.158), 13.4% of LOs open with a Tier 0 verb. The UCD D1 mean is substantially below the five-university mean excluding UCD (1.695), which may reflect formatting artefacts in UCD's module descriptor export — specifically, a 31.2% rate of null or near-null LO fields — as well as genuine quality variation; this is flagged for follow-up in the IRR phase (D4–D6 LLM scoring). The dominant Tier 0 verb across all universities is *understand* (appearing in 2,065 UCD LOs alone); at TUs, *understand* is present but is not the dominant Tier 0 form.

The TU advantage on verb observability may reflect the vocational and professional orientation of TU programmes, in which the performative specification of outcomes — what the graduate can *do in a professional context* — is more deeply embedded in curriculum culture (Department of Further and Higher Education, Research, Innovation

and Science, 2022). It may also reflect the influence of QQI programme validation processes on TU module design, or historical differences in how Bloom's taxonomy has been adopted and enforced in curriculum approval workflows across sector types.

5.2 Dominant Tier 0 Verbs

Table 6 lists the fifteen most frequent Tier 0 verbs across the full corpus. The dominance of *understand* is consistent with prior findings in the international literature (Kennedy, 2007); it is sufficiently common to suggest institutional habit rather than individual authorial choice.

Table 6: Top 15 Tier 0 verbs (full corpus, n = 226,295 LOs).

Verb	**Count**	**% of Tier 0**
understand	**[TODO: n]**	**[TODO: %]**
appreciate	**[TODO: n]**	**[TODO: %]**
know	**[TODO: n]**	**[TODO: %]**
be aware of	**[TODO: n]**	**[TODO: %]**
learn	**[TODO: n]**	**[TODO: %]**
gain	**[TODO: n]**	**[TODO: %]**
be familiar with	**[TODO: n]**	**[TODO: %]**
comprehend	**[TODO: n]**	**[TODO: %]**
recognise	**[TODO: n]**	**[TODO: %]**
grasp	**[TODO: n]**	**[TODO: %]**
explore	**[TODO: n]**	**[TODO: %]**
familiarise	**[TODO: n]**	**[TODO: %]**
enhance	**[TODO: n]**	**[TODO: %]**
cover	**[TODO: n]**	**[TODO: %]**
develop awareness	**[TODO: n]**	**[TODO: %]**

[Note: Run verb frequency query across full corpus to populate Table 3.]

5.3 D2 and D3: Behavioural Singularity and Student-Centredness

Across the full corpus, the majority of LOs demonstrate adequate student-centredness (D3 ≥ 2; **[TODO: n]**%), reflecting widespread adoption of the standard *"On completion of this module the student will be able to…"* framing. However, behavioural singularity (D2) reveals meaningful variation: **[TODO: n]**% of LOs contain two or more distinct action verbs in a single statement, and **[TODO: n]**% contain three or more. Compound LOs of this kind resist reliable assessment design and create ambiguity about which competence is being evaluated.

5.4 Institutional Format: The TU Dublin Finding

TU Dublin presented a methodologically distinct case. Scraping of 5,449 publicly accessible module catalogue pages (TUD's *Vault* system) revealed that TUD does not publish student-centred learning outcomes on its public module pages. Instead, modules are described using a module aims format: *"The aim of this module is to [verb]…"*. This format is grammatically and epistemologically distinct from an LO: it describes the module's instructional intention rather than the student's expected achievement.

Of 5,449 pages queried, 2,687 (49.3%) yielded extractable aim statements; 89 contained explicit *"students will be able to…"* LOs. The mean D1 score for extracted TUD aims (1.458) is lower than that of any other TU in the corpus and is comparable to the university cluster. However, direct comparison with institutions publishing student-centred LOs is methodologically inappropriate; the aims format is itself the finding.

A supplementary investigation discovered that TU Dublin does publish full QQI-compliant module catalogues in PDF form for at least one school. Four PDF module catalogues for the *Spatial Planning and Environmental Management* programme (TU835, Stages 1–4, 2024–25) were retrieved from the TU Dublin public website and parsed using a `pdfplumber`-based extractor targeting the Akari M1 template format. Subsequent discovery identified a second PDF format — TU Dublin's Coursebuilder (Akari M2) template — used for five further programmes. Systematic URL probing of all 26 TUD school media directories against 542 programme codes (147 from the CAO handbook plus codes identified via the TU Dublin A-Z courses page and a supplementary sweep of the TU001–TU199 short-code

range) yielded PDFs for six programmes in total:

- **TU835** (Architecture/Spatial Planning, 4 stages): 35 modules, 206 LOs in M1 template format
- **TU874** (Mathematical Sciences, 4 stages): 61 modules, 362 LOs in Coursebuilder format
- **TU078** (Mathematical Sciences postgraduate, 1 stage): 8 modules, 59 LOs in Coursebuilder format
- **TU079** (Mathematical Sciences postgraduate, 1 stage): 8 modules, 56 LOs in Coursebuilder format
- **TU248**: 4 modules, 27 LOs in Coursebuilder format
- **TU5294**: 4 modules, 26 LOs in Coursebuilder format

Combined, the PDF corpus yields 120 unique modules and 736 student-centred LOs with a mean D1 score of 1.837 — substantially higher than the Vault aims corpus and consistent with TU-sector norms. These records are included in the corpus. The remaining TUD schools do not appear to have published equivalent PDFs through discoverable URL patterns; contact with the TUD Quality Office for comprehensive coverage is recommended for future work. TUD is, to our knowledge, the only Irish HEI whose primary public module catalogue does not publish student-centred learning outcomes system-wide, representing a transparency concern independent of what may exist in internal programme documentation.

5.5 Specialist Sector: NCAD

The National College of Art and Design (NCAD) presents a methodologically and substantively distinctive case. Module descriptors were retrieved from NCAD's public website as individual PDF files and parsed to extract LO text, assessment components, and LO-to-assessment mappings. The resulting corpus (79 modules, 269 LOs) is small relative to institutional HEI corpora but constitutes the complete publicly available set for the undergraduate programmes surveyed.

NCAD returns the highest D1 mean in the corpus (1.796), with Tier 0 at 2.6% — lower than any other institution. This is consistent with the performative nature of art and design education: module LOs are characterised by observable, practice-based verbs (*demonstrate, design, develop, create, produce, exhibit*) that map naturally to high D1 tiers. The absence of Tier 0 language (*understand, appreciate*) is itself a disciplinary marker.

Equally notable is NCAD's assessment profile: across all 79 modules, zero formal examinations are recorded. Assessment is exclusively portfolio-, project-, and presentation-based — a complete CA model. This represents a natural point of comparison with the confirmed Type A modules identified at MTU, SETU, and NCI, where high-demand LOs are paired with examination-heavy assessment; at NCAD, the assessment modality is structurally aligned to the LO language by institutional design.

5.6 Private Sector: Griffith College Dublin

Griffith College Dublin (GC) is a designated higher education institution operating under the Qualifications and Quality Assurance (Education and Training) Act 2012. Module LO data were retrieved from PDF module descriptor documents linked from programme pages on the Griffith College website. Seventy-four unique PDFs were identified and downloaded; 63 (85.1%) contained parseable numbered learning outcome lists. The remaining 11 PDFs provided aims-only or empty descriptor text and were excluded.

PDFs follow a standardised format that includes module title, ECTS credits, NFQ level, numbered LOs (using Arabic or Roman numerals across two format variants), and a "Module Objectives" or "Module aims and objectives" section used as the descriptor field. No module codes are published; GC records carry the module title as the primary identifier.

The 63 modules yield 441 student-centred LOs — a mean of 7.0 LOs per module, higher than the corpus-wide average. GC returns the highest D1 mean in the corpus (1.914), with Tier 0 at 0.5% (2 LOs), the second-lowest Tier 0 rate after MIE. The verb profile is markedly action-oriented; modal verbs (*apply, analyse, evaluate, design, implement*) dominate, consistent with a strong professional and applied orientation across Griffith's programme portfolio. The distribution of assessment data was not available in the PDF format and GC contributes to LO quality analysis only.

5.7 College of Education Sector: MIE

Marino Institute of Education (MIE) contributes the smallest and methodologically most distinctive sub-corpus in this study. Four undergraduate programme handbooks were parsed: B.Ed. (Primary), B.Sc. Early Childhood Education, B.Sc. Education Studies, and B.Oid. (Irish-medium B.Ed.). The B.Oid. handbook is in Irish (Gaeilge) and is excluded from D1 scoring, as the verb lexicon is English-only; its modules are retained in the corpus with a `language=ga` flag.

Of 163 modules across the four programmes, only 14 (8.6%) contain student-centred learning outcomes meeting the trigger criterion (*On/Upon successful completion of this/the module*, or the Irish equivalent *Ar chríochnú an mhodúil*); the remaining 149 modules (91.4%) carry module descriptions but no formal LOs. B.Sc. Education Studies presents zero LO-bearing modules across its entire programme. The 49 scoreable LOs yield a D1 mean of 1.592 with Tier 0 at 0.0%.

The headline LO-absence rate is less informative than the within-institution pattern: modules *within the same programme and academic year* differ in whether they carry formal LOs, with no apparent institutional standard governing which module types are required to do so. This is not a case of a sector-wide convention against LO publication (such as might partially explain TUD's aims-format corpus); it is an inconsistency of practice within a single institution. For a teacher education college whose graduates will be expected to write LOs for their own classroom teaching, the absence of a consistent institutional LO framework is itself a notable finding. Tier 0 at 0.0% — the lowest in the corpus — suggests that where LOs *are* written at MIE, verb choice is consistently observational; the quality deficit lies in coverage, not in the language used where LOs exist.

5.8 Constructive Alignment: Maynooth University

At MU, 2,139 modules had complete D1 scoring and published assessment weight data. Cross-tabulating D1 mean tier against examination weight revealed 73 modules with mean D1 ≥ 2.0 and examination weight $\geq 70\%$. Applying a third filter — zero reported laboratory or practical hours in the timetabled T&L activities — reduced this to **49 confirmed Type A** modules: modules in which high-demand LOs are promised, examination-heavy

assessment is applied, and no practical learning activity is recorded that might provide an alternative alignment pathway.

The remaining 24 modules (high D1, exam $\geq$ 70%, but with laboratory hours present) were classified as ambiguous: the CA component might rescue alignment if the CA is authentically aligned to the high-demand LOs, but MU's data recording practice — CA as a single aggregate percentage weight, without component type — does not permit this determination from data alone.

5.9 Constructive Alignment: Munster Technological University

MTU's Akari assessment management system enabled a substantially more precise analysis. Of 4,987 modules with both D1 scoring and confirmed assessment data, **114** met the Type A threshold (mean D1 $\geq$ 2.0, exam $\geq$ 70%). The LO-to-component index in Akari maps each assessment component to the specific LO indices it is designed to assess, enabling the following classification:

- **55 confirmed misaligned**: high-demand LOs (D1 $\geq$ 2) are specifically mapped to the formal examination component, with no CA component also targeting those LOs. The examination is the sole instrument for assessing the highest-demand outcomes.
- **101 partially rescued**: the module is examination-heavy overall, but high-demand LOs are also mapped to at least one CA component. The headline exam percentage is misleading as a misalignment indicator; the internal LO mapping reveals adequate alignment.
- **1 fully aligned**: all high-demand LOs are mapped exclusively to CA components despite the high examination weight.
- **7**: LO index data insufficient to classify.

The distribution of CA types in confirmed misaligned MTU modules is revealing (Table 7). Short Answer Questions (SAQ) is the most frequent CA type (20 instances), which is effectively examination-like in cognitive structure: it assesses recall and short-form reproduction rather than the higher-order performance implied by the high-demand LO verbs.

Table 7: CA component types in MTU confirmed misaligned modules ($n = 55$).

CA Component Type	Count
Short Answer Questions	20
Other	14
Written Report	12
Project	12
Practical/Skills Evaluation	10
Essay	7
Open-book Examination	5
Presentation	4
Non-MTU Exam	1

5.10 Constructive Alignment: South East Technological University

SETU's Akari data provides a third instance of the gold-standard LO-to-component analysis. Of 3,772 modules with D1 scoring and confirmed assessment data, **7** meet the confirmed Type A threshold (mean D1 ≥ 2.0, exam $\geq 70\%$, high-demand LOs mapped exclusively to the examination component), with a further **40** partially rescued. SETU also yields **5 confirmed Type B** modules — low D1 (mean < 1.5) paired with CA-heavy assessment — representing the converse alignment problem: assessment that exceeds the aspiration of the LO language. The CA-only component type in SETU confirmed misaligned modules is uniformly Assignment, consistent with the observation that written assignments are the most common CA rescue mechanism across Akari institutions.

The lower absolute count of Type A modules at SETU relative to MTU (7 vs 55) is partially attributable to SETU's lower overall D1 mean (1.529 vs 1.846): fewer modules clear the D1 ≥ 2.0 threshold. This raises the question of whether a lower Type A count represents better alignment or simply lower LO ambition.

5.11 Constructive Alignment: National College of Ireland

NCI's publicly accessible Akari instance provides a fourth gold-standard dataset. Of 749 modules analysed, **2** are confirmed Type A and **26** partially rescued. No Type B modules were identified. NCI's D1 mean (1.627) is comparable to the university cluster, despite NCI's classification as a private/specialised institution; its Tier 0 rate (10.4%) is the highest in the non-university sample.

Table 8 summarises the alignment findings across all four institutions.

Table 8: Constructive alignment findings by institution and data method.

	MU	**MTU**	**SETU**	**NCI**
Modules with D1 + assessment data	2,139	4,987	3,772	749
Type A candidates (D1 $\geq$ 2, exam $\geq$ 70%)	73	114	47	28
Confirmed misaligned	49[a]	55[b]	7[b]	2[b]
Partially rescued	24[c]	101	40	26
Type B (low D1, CA-heavy)	n/a	—	5	0
Method	Proxy	LO-index	LO-index	LO-index
CA type data available	No	Yes	Yes	Yes

[a] Confirmed by lab-hours proxy (zero practical hours). [b] Confirmed by LO-to-component index mapping. [c] Ambiguous: exam $\geq$ 70% but lab hours present; CA may rescue alignment.

The methodological contrast between MU (weight-only + lab-hours proxy) and the three Akari institutions (MTU, SETU, NCI) is itself a substantive finding. The 24 MU modules classified as ambiguous would be resolvable with component-level CA recording. The 101 MTU modules that appear misaligned on exam-weight alone but are resolved as partially aligned by the LO index are modules that a weight-only analysis would misclassify. The precision of the Akari data is a direct consequence of the system's design; it suggests that the data infrastructure of assessment management systems has a direct bearing on the quality of evidence available for institutional review. The confirmed misalignment count across all four institutions (113 confirmed Type A modules) represents a conservative lower bound; the five institutions with weight-only or no assessment data would yield additional

cases under the same criteria.

5.12 Readability of Module Descriptors and Learning Outcomes

[Note: This section reports planned readability analysis on module descriptor text and LO statements across the full corpus. No API cost — pure text processing.]

A learning outcome that specifies an observable behaviour in the correct grammatical form may nonetheless be inaccessible to the student it is written for. Verb observability (D1) captures the *epistemic* quality of an LO statement; its *communicative* accessibility is a distinct dimension. Module descriptors and LO text are public-facing documents: students consult them when choosing programmes, planning their studies, and understanding what will be assessed. Their readability is therefore a student equity concern as much as a quality assurance one.

We apply a suite of established readability measures to module descriptor text and LO statements across all institutions in the corpus:

- **Flesch-Kincaid Grade Level (FKGL)**: estimates the US school grade level required to comprehend a text; based on sentence length and syllable count. A FKGL of 12–14 is typical for HE module descriptors; values above 16 suggest prose that may impede student comprehension.
- **Flesch Reading Ease (FRE)**: the inverse of FKGL on a 0–100 scale (higher = more readable). More intuitive for non-specialist audiences and policy reporting.
- **Gunning Fog Index**: penalises words of three or more syllables, directly flagging the nominalisation problem identified in systemic functional linguistics (SFL) accounts of academic register (Halliday, 1993). A Gunning Fog score above 12 is considered difficult; above 17, obscure.
- **Lexical density**: the ratio of content words (nouns, verbs, adjectives, adverbs) to total words. In SFL, high lexical density characterises written, formal register; very high density in module descriptors may indicate information packing that reduces accessibility without increasing clarity (Halliday, 1989).

- **Average sentence length**: a direct measure of syntactic complexity; correlates with cognitive load independent of vocabulary.

Table 9 reports preliminary findings from the two institutions for which the full pipeline has been run to date (MU and MTU; $n = 2{,}779$ and $n = 6{,}110$ modules respectively with sufficient descriptor text).

Table 9: Readability of module descriptors: MU (PU) vs MTU (TU). Higher FKGL and FOG indicate less readable text; higher FRE indicates greater readability; LD = lexical density (%).

Measure	MU ($n = 2{,}779$)		MTU ($n = 6{,}110$)	
	Mean	Median	Mean	Median
FKGL	18.74	17.53	20.14	17.82
FRE	12.25	7.35	6.39	0.00
Gunning Fog	21.35	20.12	22.75	20.44
Lexical Density (%)	63.4	62.2	69.4	69.2
Avg Sentence Length	25.2	21.8	23.4	16.0

Both institutions produce module descriptors that are substantially harder to read than is desirable for public-facing student documents. Mean FKGL values of 18.7 (MU) and 20.1 (MTU) correspond to postgraduate reading-level requirements; the Gunning Fog scores (21.4 and 22.7 respectively) far exceed the threshold of 17 typically considered obscure. MTU descriptors show markedly higher lexical density than MU (69.4% vs 63.4%), consistent with the SFL prediction that professional/technical registers sustain higher information packing (Halliday, 1989) — a finding that warrants attention given the TU sector's stated commitment to student-centred learning.

We computed Pearson correlations between D1 verb tier (per module mean) and each readability measure. The only substantive correlation was between D1 and lexical density (MU: $r = 0.225$; MTU: $r = 0.191$), suggesting that modules specifying more observable behaviours tend to use slightly denser, more content-rich prose — consistent with the hypothesis that precise LO writing requires richer noun-phrase structures to specify conditions and criteria. Correlations between D1 and sentence- level measures (FKGL, FOG, ASL) were negligible in both institutions ($|r| < 0.07$), indicating that verb vagueness

and syntactic complexity vary largely independently: a module can be written in plain sentences yet specify unobservable outcomes, or use complex prose to specify measurable ones.

[TODO: Extend readability pipeline to all nine corpus institutions (DCU, Galway, UCC, UCD, UL, ATU, SETU, TUD) once data collection is complete. Add cross-tab by NFQ level and faculty/discipline. Consult Susan Gottlöber re Humanities descriptor language norms; Niamh Fortune re Social Sciences.]

A note on the shadow curriculum

The analyses in Sections 5.13–5.15 share a methodological caveat that is also a policy argument. Module descriptor text and published LOs constitute the *visible curriculum*: the documented commitment made to students, accreditation bodies, and the public. They do not constitute the *enacted curriculum*. UDL practices, skills development activities, and data literacy components that are present in teaching and assessment but absent from documentation are invisible to this analysis — and to every analysis dependent on public-facing records.

This is not merely a measurement limitation. Good practice that goes undocumented cannot be credited in national benchmarking, replicated by colleagues, or communicated to prospective students making module choices. Where the analyses below identify weak signal for UDL, skills, or data literacy, the most accurate reading is: *the documentation does not claim it*. Whether the practice exists in the shadow curriculum is a question this dataset cannot answer — but it is one that institutions are well-placed to investigate and, where the evidence supports it, to document visibly. The message is not deficit but opportunity: if it is happening, tell everyone.

Where assessment component data is available (MTU, SETU, DkIT, NCI via Akari; MU via CA/exam weighting), it provides a second window onto enacted practice that the LO text alone cannot open. Sections below flag cases where assessment data corroborates or contradicts the UDL, skills, or alignment signal in the descriptor text.

5.13 Universal Design for Learning: Proxy Analysis

[Note: This section reports preliminary UDL proxy findings for MU (2,139 modules). Extension to the full corpus is planned. ***[PLACEHOLDER: narrative to be developed with Niamh Fortune and Susan Gottlöber.]]***

Universal Design for Learning (UDL) (CAST, 2018) provides a complementary lens to verb-level quality analysis. Where D1 asks what cognitive performance an LO specifies, UDL principles ask whether the module's design creates the conditions for diverse learners to achieve that performance. We operationalised three UDL principles as proxies from module descriptor data at MU:

- **Representation** (Principle I): diversity of teaching and learning modalities recorded in the timetabled activity (lectures, laboratory, tutorial, planned self-directed learning).
- **Action and Expression** (Principle II): diversity of assessment component types across the module.
- **Engagement** (Principle III): presence of engagement-signalling keywords in module descriptor text (e.g., *collaborative*, *problem-based*, *authentic*, *reflective*).

Across 2,139 MU modules with complete data, preliminary findings indicate:

- 40.3% of modules report a single teaching and learning modality (lectures only), indicating low UDL Representation signal.
- Only 5.4% of modules show strong signal across all three UDL principles.
- 17.9% show no UDL signal (single T&L modality, no engagement keywords, single assessment component).
- A weak positive correlation exists between UDL composite score and D1 mean ($r = 0.059$); engagement keyword presence shows the strongest relationship with verb observability ($r = 0.085$).
- Department-level variation is pronounced: modules in Critical Skills show the highest combined UDL and D1 profiles; modules in History show the lowest on both dimensions.

These findings are preliminary and subject to the limitations of keyword-based proxy analysis. Descriptor text quality and completeness varies across departments and individuals. **[TODO: Extend UDL analysis to full nine-institution corpus once descriptor text is normalised across institutions. Consult with Niamh Fortune and Susan Gottlöber re UDL in Irish HE context (Froebel/Social Sciences and Humanities perspectives respectively). Consider whether Strand 4 UDL paper material should be partially incorporated here or remain a separate publication.]**

5.14 HEA Data and Digital Skills: Module Tagging Case Study

*[Note: **[PLACEHOLDER: data and narrative to be developed with Tim Thompson (VP for Students and Learning) and John G. Keating.]**]*

In response to the HEA's *Human Capital Initiative* mandate, Maynooth University undertook a systematic tagging of modules contributing to Data and Digital Skills competencies across its curriculum. This exercise combined human expert tagging with AI-assisted classification, enabling an agreement analysis between human and algorithmic judgements.

[TODO: Develop this section once John shares HEA Data/Digital reports. Key content to include: (a) methodology for human + AI tagging; (b) agreement analysis results; (c) distribution of Data/Digital modules across faculties and NFQ levels; (d) connection to LO quality findings — does higher D1 verb tier correlate with stronger Data/Digital signal? (e) relevance to national skills policy. This section connects Strands 3 and 5 of the broader research programme.]

5.15 Graduate Skills Alignment: QQI Framework

*[Note: **[PLACEHOLDER: analysis complete; narrative to be developed with Máire Buckley, Experiential Learning Officer, MU.]**]*

QQI's *Graduate Attributes Framework* (Quality and Qualifications Ireland, 2022b) identifies a set of capabilities expected of Irish higher education graduates, including critical thinking and problem solving, communication, collaboration, creativity and innovation, research and inquiry, ethical responsibility, and self-direction. These are not discipline-specific but

constitute a cross-sectoral baseline against which programme and module documentation can be examined.

We operationalise each attribute as a keyword and phrase set applied to both LO text and module descriptor text across all fourteen institutions. Because documentation is the unit of analysis (see shadow curriculum note above), the measure captures whether institutions *claim* these attributes in their public module records, not whether they deliver them.

Table 10: QQI Graduate Attribute keyword targets for corpus scanning. Sources: LO text and module descriptor text; assessment component labels where Akari data is available (MTU, SETU, DkIT, NCI, MU).

Attribute	**LO / Descriptor targets**	**Assessment signal**
Critical thinking	evaluate, critique, analyse, justify, argue, assess	essay, viva
Communication	present, report, communicate, write, articulate	presentation, report
Collaboration	collaborate, group work, team, peer review	group project
Creativity & innovation	design, create, develop, innovate, generate	portfolio, project
Research & inquiry	investigate, research, evidence-based, literature	dissertation, project
Ethical responsibility	ethical, responsible, sustainability, equity, justice	reflection
Self-direction	self-directed, independently, manage own learning	portfolio

[TODO: Run scan-skills.mjs across all 14 corpus institutions. Produce: (a) per-institution attribute coverage rates (proportion of modules with at least one keyword hit per attribute); (b) sector-level comparison (PU vs TU vs IT vs CE vs PS); (c) cross-tab with D1 tier (do high-D1 modules also show stronger skills signal?); (d) where Akari assessment data is available, flag modules where assessment component type corroborates the skills claim in the LO text. Consult Máire Buckley re QQI attribute definitions and any

MU-specific graduate attribute policy. Flag whether Data/Digital overlaps with any QQI attribute (likely: research and inquiry, self-direction) to avoid double-counting with Section 5.14.]

5.16 Alignment with National Policy Standards

*[Note: **[PLACEHOLDER: analysis complete; narrative and policy interpretation to be developed with Lisa O'Regan, Centre for Teaching & Learning, MU. See `papers/policy-landscape-reference.md` for full policy language and mapping table.]]***

The three-level policy framework described in Section 2.2 establishes concrete, testable requirements for LO quality. Table 11 maps these requirements to the D-dimensions of our rubric and summarises the corpus-level compliance picture.

Table 11: Mapping of policy requirements to rubric dimensions and corpus findings. Sources: ESG 2015 Standard 1.2; QQI Validation Policy 2021; NFQ Grid of Level Indicators; Dublin Descriptors (Cycle 1 and 2).

Policy requirement	**Rubric dimension**	**Corpus finding**
"Specific and sufficiently detailed" (QQI)	D1 Verb observability	10.4% of corpus LOs use Tier 0 (unobservable) verbs; range 2.6%–23.9% by institution
"Specific" — one outcome per statement (implied)	D2 Behavioural singularity	**[TODO: D2 mean scores by institution — to insert]**
Student-centred LO format (ESG 1.2; Bologna)	D3 Student-centredness	TUD Vault (2,791 LOs) uses aims format; 736 student-centred LOs recovered from PDF catalogues across 6 programmes (TU835, TU874, TU078, TU079, TU248, TU5294); MIE 91.4% of modules carry no student-centred LOs
"Appropriate to specified NFQ level" (QQI)	D6 NFQ level calibration (LLM)	**[TODO: D6 results — to insert once LLM scoring complete]**
Assessment ensures MIPLOs are assessed (QQI)	Constructive alignment analysis	123 confirmed Type A modules across 5 institutions; MU: 49; MTU: 55; SETU: 7; DkIT: 10; NCI: 2
LOs present for all modules (QQI institutional QA)	LO publication rate	MIE: 8.6% module LO rate; UCD: 31.2% null; UL: 37.8% null; TUD: aims-only

Several findings warrant particular attention from a policy compliance perspective. First, the prevalence of Tier 0 verbs (*understand*, *know*, *appreciate*) across all fourteen institutions indicates that despite three decades of outcomes-based education policy, unobservable

language remains embedded in Irish HE module documentation at scale. Second, the MIE no-LO rate (91.4%) and TUD aims-format corpus represent structural rather than quality issues: the problem is not poorly written LOs but the absence of the student-centred outcome format that QQI and the Bologna process require. Third, the 123 confirmed Type A modules identified across five institutions represent documented cases in which QQI's assessment-alignment requirement is not met: high-demand LOs are promised but examination-only assessment is applied, producing a verifiable gap between the MIPLO commitment and the assessment instrument designed to verify it.

[TODO: Lisa O'Regan to develop: (a) contextualise findings against QQI's own review/audit findings where publicly available; (b) consider whether the null LO rates at UCD/UL represent institutional QA process failures or publication system artefacts (and the policy implications of each); (c) institutional response framing — recommendations for QA process improvement at programme approval stage; (d) international comparator — how does Ireland compare with EHEA peers on LO quality? Consider referencing EURASHE / EUA Learning and Teaching Thematic Peer Group findings if available.]

6 Discussion

6.1 The PU–TU Gap

The consistent D1 advantage of the TU sector over the public university cluster — $\Delta = 0.203$ (PU mean 1.572, TU mean 1.775), reproduced in direction across every institution in both clusters — merits careful interpretation. It would be tempting to read this as evidence of a quality differential favouring TUs, and in the specific dimension of verb observability this reading is defensible. However, the interpretation should be contextualised.

TU programmes, rooted in applied and professional education, are designed around the demonstration of competence in specific contexts. The programme approval and review culture in the technological sector, historically regulated through QQI processes with strong emphasis on measurable and vocational outcomes, may have produced a curriculum documentation practice in which observable verbs are more consistently selected. This is

not necessarily evidence of deeper constructive alignment; it may reflect a discipline-level norm about how outcomes are articulated.

Universities, by contrast, house a wider range of disciplines including humanities, social sciences, and pure sciences in which the objectives of education are genuinely more difficult to reduce to observable performance. A module in philosophy, history, or theoretical mathematics may legitimately aim for transformative *understanding* that resists specification in the terms that Bloom's taxonomy most easily accommodates. The persistent presence of *understand* in university LOs may therefore reflect both a quality problem (weak outcome design) and an epistemological resistance (the genuine difficulty of specifying what excellent philosophical or historical reasoning looks like in verb form).

The collaboration of Maynooth University colleagues in the authorship of this paper reflects this tension directly. Faculty of Arts and Humanities and Faculty of Social Sciences perspectives are essential for distinguishing the legitimate complexity of humanistic learning objectives from the habitual use of vague language that could, with appropriate design support, be made more precise without sacrificing disciplinary integrity.

6.2 Constructive Alignment: A Solvable Problem

The finding that 113 modules across four institutions exhibit confirmed constructive misalignment (MU: 49; MTU: 55; SETU: 7; NCI: 2) is notable not primarily because of its scale, but because of what it implies about current quality assurance processes. These are not hidden or theoretical misalignments: they are encoded in publicly accessible module descriptors and institutional assessment records. A module approval process with routine access to D1 verb scoring and assessment modality data would, in principle, detect and flag these modules before students enrol.

The 101 MTU modules initially appearing misaligned but resolved by LO-index mapping makes the complementary point: apparent misalignment is not always actual misalignment. Surface-level analysis of exam percentages without reference to the LO-level mapping of assessment components produces both false positives (modules flagged as misaligned that are in fact well-designed) and false negatives (modules that appear acceptable because their CA weight is high but whose CA is examination-like in structure, as the SAQ finding suggests).

6.3 Policy Implications

Three policy implications follow from these findings.

First, module approval processes should incorporate routine LO quality checking at the verb level. This does not require human expert review of 232,000 LOs; it requires a lightweight automated scoring step integrated into module registration or approval systems. Our pipeline, which processes the full corpus in under an hour on standard hardware, is deployable at institutional scale.

Second, assessment data recording standards should be revised to capture component type, not merely percentage weight. The ambiguous category in MU's analysis — 24 modules that cannot be resolved without component-type data — is an artefact of data recording practice, not of the underlying educational reality. The investment required to capture *Project*, *Short Answer Questions*, *Presentation* alongside the percentage weight is minimal; the quality assurance value is substantial.

Third, the TU Dublin finding invites a national conversation about the publication of learning outcomes on public module catalogue pages. If module descriptors are the primary instrument through which students make informed programme choices, and through which the sector accounts for what it teaches, then the widespread non-publication of student-centred learning outcomes is a transparency issue that QQI's review processes should address directly.

6.4 Limitations

Several limitations should be noted. The D1–D3 dimensions capture syntactic and lexical features of LO text; they do not assess the underlying quality of educational design or the lived experience of alignment in the classroom. A module with perfectly observable, student-centred, singularly specified LOs may still be poorly taught or poorly assessed in practice; conversely, a module with Tier 0 verbs in its formal descriptor may be taught with rigour and assessed authentically by a skilled practitioner.

The lexicon-based approach to D1 scoring is limited by lexicon coverage and by the context-dependence of verb interpretation. *Review*, for example, is assigned Tier 1 in our lexicon, but in some disciplinary contexts it describes a complex evaluative act more appropriately

assigned to Tier 2. The v1.3 lexicon includes a context-dependent verb register to flag such cases for manual review; a fully automated treatment of context-dependence awaits LLM-based scoring (D4–D6, in progress).

The constructive alignment analysis at MU is limited by the weight-only recording of continuous assessment. The 24 ambiguous modules cannot be resolved without component-type data, and some proportion of the 49 confirmed Type A modules may contain CA components not captured in the descriptor that do provide alignment rescue. Similarly, the MTU analysis is limited to modules for which the Akari system records LO-to-component mappings; a subset of modules had no such mapping and were excluded.

7 Conclusion and Recommendations

This study provides the first cross-institutional, corpus-scale characterisation of learning outcome language quality in Irish higher education. Across 226,295 LOs from nine institutions, we find a consistent gap in verb observability between university and technological university sectors, with universities showing higher rates of unobservable Tier 0 verbs and lower mean D1 scores. The gap is reproducible across all institutions within each cluster and is unlikely to be artefactual.

The extension to constructive alignment reveals that verb quality is necessary but not sufficient: 55 MTU modules (and a minimum of 49 MU modules) exhibit confirmed misalignment between high-demand LO language and assessment modalities that cannot elicit the specified performance. The methodological contrast between MU and MTU demonstrates that the precision of this analysis is directly determined by the granularity of assessment data recording practice.

We offer the following recommendations:

1. **Integrate automated LO quality scoring into module approval workflows.** Lightweight rule-based scoring of the leading action verb is sufficient to flag Tier 0 usage for author review. This does not constrain academic freedom; it surfaces language choices for deliberate consideration.

2. **Revise assessment data recording standards to capture component type.**

The distinction between *Project* and *Short Answer Questions* as CA modalities is not a bureaucratic nicety; it is the difference between a resolvable and an unresolvable alignment question.

3. **Establish a national expectation for LO publication on public module catalogue pages.** The TU Dublin finding indicates that the absence of student-centred LOs on public pages is not inevitable; it is an institutional choice. QQI programme review should include a check on public LO publication.

4. **Extend the analysis to D4–D6.** Rule-based scoring captures syntax; LLM-based scoring of scope, assessability, and NFQ calibration will extend the analytical depth and move toward a composite quality score suitable for institutional dashboard reporting.

5. **Involve academic development units in disciplinary interpretation.** The university–TU gap cannot be addressed by blanket verb-replacement advice. Humanities and social science disciplines require disciplinary-appropriate guidance on how to specify complex understanding in assessable terms without reducing learning to performance. Faculty-level academic development, informed by discipline-specific examples from the corpus, offers the most promising pathway.

The data, verb lexicon, and scoring pipeline are available to institutional partners and QQI on request. **[TODO: Add data availability / supplementary materials statement.]**

Acknowledgements

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A LLM Evaluator Prompts

This appendix reproduces the exact prompts used in the policy-grounded LLM evaluation (`score-llm-v2.mjs`). Full scripts and all sixteen institutional context blocks are available in the project repository (see Section 7).

A.1 System Prompt: Policy Layer and Evaluation Rubric

The system prompt is fixed across all API calls and contains the complete policy grounding and rubric (approximately 1,200 tokens; amortised across N LOs per call — see Table 4).

```
You are an expert evaluator of learning outcomes (LOs) in Irish higher
education. Your evaluations are grounded in the following policy framework,
which defines what a well-formed LO must achieve in this jurisdiction.

## QQI Policies and Criteria for the Validation of Programmes (QP.17, 2017)
- Module LOs must be "explicitly specified" - vague or generalised
  statements are non-compliant.
- LOs must not be "vague, ambiguous or generalised."
- LOs must be "appropriate to award title and credit volumes."
- Minimum Intended Module Learning Outcomes (MIMLOs) are the binding
  specification.

## QQI Core Policies and Criteria (QP 24, 2024)
- LOs must be plainly written so that students can understand what they
  are expected to achieve.
- QQI's stated normative reference for LO writing guidance is Cedefop (2017).
```

ECTS Users' Guide (European Commission, 2015)

- LOs describe what the learner will be able to know, understand, and do after completing a learning activity.
- LOs must describe demonstrable, observable achievement - not curriculum content coverage.
- Verbs such as "understand" and "know" are explicitly discouraged as unobservable.
- One ECTS credit corresponds to 25-30 hours of student workload - scope must be calibrated accordingly.

Cedefop (2017) - Defining, Writing and Applying Learning Outcomes (QQI's normative reference)

- Each LO should specify a single observable behaviour using an active verb.
- Verbs should be drawn from a recognised taxonomy (e.g. Bloom's) appropriate to the cognitive level.
- LOs should be specific, measurable, and student-centred.
- Common errors: using vague verbs (know, understand, appreciate, be familiar with); writing aims or curriculum content as LOs; writing teacher-centred rather than learner-centred statements.

NFQ Level Descriptors

NFQ Level 6: Knowledge - broad general knowledge of field; Skill - demonstrate a range of standard skills; Competence - act independently in familiar contexts, limited autonomy.

NFQ Level 7: Knowledge - broad knowledge with some theoretical depth; Skill - apply range of skills to known or new contexts; Competence - some autonomy in decision-making, take responsibility for outcomes.

NFQ Level 8 (Honours Bachelor): Knowledge - detailed theoretical and conceptual knowledge; Skill - demonstrate advanced skills including synthesis, evaluation, critical analysis; Competence - act with autonomy in novel complex contexts, take initiative.

NFQ Level 9 (Masters): Knowledge - specialised knowledge at forefront of field, including research; Skill - highly developed skills in research, analysis, and problem-solving in unpredictable contexts;

Competence - act with substantial autonomy, manage complex professional activities.

NFQ Level 10 (Doctoral): Knowledge - original contribution to knowledge at forefront of field; Skill - expert research and advanced technical skills; Competence - exercise substantial authority, innovation, full autonomy in academic or professional activity.

Evaluation Rubric

D1 - VERB OBSERVABILITY TIER (for IRR against NLP pipeline)

0 = Tier 0: Non-observable verb (know, understand, appreciate, be aware of, be familiar with, learn) - cannot be assessed as stated.

1 = Tier 1: Low-demand observable verb (identify, define, describe, list, name, state, recall) - assessable but cognitively undemanding.

2 = Tier 2: Mid-demand observable verb (apply, analyse, explain, compare, demonstrate, calculate, classify) - standard HE-level demand.

3 = Tier 3: High-demand observable verb (evaluate, critique, synthesise, design, create, construct, formulate, justify) - advanced demand.

D4 - SCOPE APPROPRIATENESS

Does the scope match the NFQ level and ECTS credit volume of the module?

0 = Scope is wrong: either a micro-task or a programme-level goal.

1 = Scope is plausible but imprecise.

2 = Scope is module-appropriate - a single assessable outcome at the right granularity for this credit weight.

3 = Scope is precisely calibrated - matches module content and credit weight exactly.

D5 - ASSESSABILITY SIGNAL

Can a specific, valid assessment be pictured for this outcome?

0 = No plausible assessment - describes a state, not a performance.

1 = An assessment can be imagined but requires significant reinterpretation.

2 = A clear assessment method is implied (exam question, lab task, report, presentation, portfolio).

```
3 = The LO directly signals a specific assessment approach, including
    context or artefact type.

D6 - NFQ LEVEL CALIBRATION
Is the cognitive demand appropriate for the stated NFQ level?
0 = Substantial mismatch - e.g. recall-only verb on NFQ 8/9.
1 = Mild mismatch - one level below or above expected cognitive demand.
2 = Consistent with NFQ level - the cognitive demand is appropriate.
3 = Precisely calibrated - verb and content complexity match both NFQ
    level and module context exactly.

Respond with ONLY valid JSON in exactly this schema:
{
  "d1_verify": { "score": <0-3>, "verb": "<extracted verb>",
                 "rationale": "<one sentence>" },
  "d4":        { "score": <0-3>, "rationale": "<one sentence>" },
  "d5":        { "score": <0-3>, "rationale": "<one sentence>" },
  "d6":        { "score": <0-3>, "rationale": "<one sentence>" }
}
```

A.2 Representative User Prompt

The user prompt is constructed per batch. The institutional context block is selected by institution code from a lookup table of sixteen entries (one per corpus institution); the module and LO fields are populated from the scored corpus record. This example shows a single-LO call for clarity; in batch mode the `LO 1`, `LO 2`, ... blocks are repeated within a single user message and the model returns a JSON array of the same length.

```
INSTITUTIONAL CONTEXT - MU:
Maynooth University: Action verb is mandatory - "the important word in
every learning outcome is the action verb." Staff required to write LOs
"appropriate to learners' level and stage." LOs should identify WHAT
the student does, not HOW. Maximum 8 LOs per module enforced by system.
```

```
LO 1 [AC205_LO1]:
Module: AC205 - MANAGEMENT & COST ACCOUNTING | NFQ Level 8 | 10 ECTS
Descriptor: The purpose of this course is to equip students with an
  explanation of management accounting as a basis for developing the
  tools and knowledge required to meet management information needs.
  Students will study the organisational content of management accounting;
  cost accumulation for inventory valuation and profit measurement;
  decision making; information for planning control and performance
  measurement, and ethics.
Text: "Describe and evaluate the management accountant's role,
  organizational environment, accounting information systems and
  possible ethical issues they face."

Evaluate D1_VERIFY, D4, D5, D6. Return JSON only.
```

A.3 Output Schema and Example Response

The model returns one JSON object per LO (or a JSON array in batch mode). Scores are clamped to $[0, 3]$ and rounded to the nearest integer. Records that fail JSON parsing or omit any required dimension are logged as errors and excluded from analysis; they are retried automatically on the next run.

Example response for the LO shown in Appendix A.2:

```
{
  "d1_verify": {
    "score": 1,
    "verb": "describe",
    "rationale": "The verb 'describe' is a low-demand observable verb,
                  making it assessable but not cognitively challenging."
  },
  "d4": {
    "score": 2,
    "rationale": "The scope is module-appropriate, addressing the role
                  of management accountants within the module context."
```

```
  },
  "d5": {
    "score": 2,
    "rationale": "A clear assessment is implied, such as an exam question
                  or report requiring description and evaluation of the
                  specified topics."
  },
  "d6": {
    "score": 1,
    "rationale": "Mild mismatch: the cognitive demand of 'describe' is
                  below what is expected for NFQ Level 8."
  }
}
```


The full set of sixteen institutional context blocks, all evaluation scripts, and the complete scored output files are available at the project repository (see Section 7).

B Human Evaluation Rating Interface

This appendix describes the browser-based rating tool used to collect human evaluation data. The tool is publicly accessible at:

https://jkeatingmu.github.io/MU-finder/slo-study/

B.1 Architecture

The application is a fully static single-page application (React/Vite), deployed to GitHub Pages. It requires no authentication, no server, and no software installation. Rating data are submitted directly from the browser to a Supabase PostgreSQL instance via the Supabase REST API, using an insert-only Row Level Security policy: anonymous users may insert rows but cannot read, update, or delete any record. The researcher retrieves accumulated ratings via authenticated access to the Supabase dashboard or by direct SQL query.

B.2 Rating Workflow

1. The rater selects an institution from the sidebar. The full LO corpus for that institution loads from a pre-scored JSON file.

2. The rater uses the search and filter controls to locate LOs of interest (by keyword, D1 tier, or NFQ level).

3. Expanding an LO card reveals the full module context, the NLP measurement scores (D1–D3), and the LLM evaluation scores ($D1_{\text{verify}}$, D4–D6), providing the rater with a reference point while maintaining independence.

4. The rater clicks *Rate this LO* to open the rating modal.

5. The modal presents the LO text at the top. A collapsible *Show module descriptor* toggle is available below the LO text for raters who need the broader module context (particularly relevant for D4–D6).

6. The modal solicits a score and optional rationale for all six dimensions D1–D6 in sequence. Scores are entered via a labelled range slider (0–3 for D1/D4/D5/D6; 1–3 for D2/D3). The rubric descriptor for the selected score updates dynamically below each slider, colour-coded red/orange/green/blue for scores 0/1/2/3. Rationale text boxes include a prompt not to enter personal information. A final *Confidence in these ratings* slider (1–5) records methodological metadata about rater certainty.

7. On submission, the record is inserted into the `lo_ratings` table with a server-side timestamp. The rater's browser records the `lo_id` in `localStorage`; rated cards display a *Rated* indicator on subsequent visits.

B.3 Data Schema

The `lo_ratings` table schema is as follows:

Column	Type	Description
`id`	UUID	Primary key (auto-generated)
`created_at`	timestamptz	Submission timestamp (server-side)
`lo_id`	text	Unique LO identifier (corpus key)
`institution`	text	Institution code (e.g. MU, DCU)
`module_code`	text	Module code (e.g. AC205)
`lo_text`	text	Full LO text (for audit)
`nfq_level`	integer	NFQ level of the module
`rater`	text	*anonymous* (pending ethics approval)
`d1`	integer [0–3]	D1 verb observability score
`d1_rationale`	text	Optional rationale
`d2`	integer [1–3]	D2 behavioural singularity score
`d2_rationale`	text	Optional rationale
`d3`	integer [1–3]	D3 student-centredness score
`d3_rationale`	text	Optional rationale
`d4`	integer [0–3]	D4 scope score
`d4_rationale`	text	Optional rationale
`d5`	integer [0–3]	D5 assessability score
`d5_rationale`	text	Optional rationale
`d6`	integer [0–3]	D6 NFQ calibration score
`d6_rationale`	text	Optional rationale
`confidence`	integer [1–5]	Rater's self-assessed confidence

Rater identity is currently recorded as *anonymous* pending institutional ethics approval for identified data collection. The schema accommodates a named `rater` field; authenticated rating (e.g. via institutional SSO or a shared session token) is a planned extension that will be activated once ethical clearance is in place.